

PU5558-PU5567-computer-lab-report

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Introduction

The following report is based on Provisional Patient Reported Outcome Measures (PROMS) specifically on the outcome of Hip and Knee Replacement procedures completed between April 2020 to March 2021, provided by NHS England. For this report the focus will be on the outcome measures of knee replacement surgery. Specific outcome measures pertaining to the Quality Of Life (QOL) to the patient have been recorded through a visual analogue scale designed to measure the self-reported status of the patient's quality of life pre/post-surgery. These values gathered from the survey will be used to inform the models.

Load necessary packages

```
library(tidyverse)
library(tidymodels)
library(corrplot)
library(psych)
library(ggplot2)
library(rmarkdown)
library(dplyr)
library(reshape2)
library(vip)
library(Hmisc)
library(VIM)
library(caret)
library(flextable)
library(gtsummary)
library(data.table)
library(randomForest)
```

Load chosen dataset

```
kneeData <- read_csv("Knee_Replacement_CCG 2021.csv")
kneeData <- kneeData %>%
  select(-Procedure, -Year, -"Revision Flag", -"Provider Code") %>%
  drop_na() %>%
  unique()
```

```
glimpse(kneeData)
```

```
## Rows: 4,528
## Columns: 77
## $ `Age Band` <chr> "*", "*", "*", "*",~
## $ Gender <chr> "*", "*", "*", "*",~
## $ `Pre-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By` <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period` <dbl> 2, 4, 4, 2, 4, 2, 2~
## $ `Pre-Op Q Previous Surgery` <dbl> 2, 1, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Disability` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Heart Disease` <dbl> 9, 9, 9, 9, 9, 1, 9~
## $ `High Bp` <dbl> 1, 9, 9, 1, 1, 1, 9~
## $ Stroke <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Circulation <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Lung Disease` <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ Diabetes <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ `Kidney Disease` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Nervous System` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Liver Disease` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Cancer <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Depression <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Arthritis <dbl> 9, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Mobility` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Self-Care` <dbl> 1, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Activity` <dbl> 3, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Discomfort` <dbl> 2, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Anxiety` <dbl> 1, 1, 2, 2, 1, 2, 1~
## $ `Pre-Op Q EQ5D Index Profile` <dbl> 21321, 21221, 21332~
## $ `Pre-Op Q EQ5D Index` <dbl> 0.364, 0.691, 0.030~
## $ `Post-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Assisted By` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Post-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Post-Op Q Disability` <dbl> 1, 2, 2, 2, 2, 1, 2~
## $ `Post-Op Q Mobility` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Self-Care` <dbl> 2, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Activity` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Discomfort` <dbl> 1, 1, 1, 1, 1, 2, 2~
## $ `Post-Op Q Anxiety` <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Satisfaction` <dbl> 2, 3, 1, 2, 1, 4, 3~
## $ `Post-Op Q Sucess` <dbl> 1, 1, 1, 1, 1, 4, 1~
## $ `Post-Op Q Allergy` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Bleeding` <dbl> 2, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Wound` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Urine` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Further Surgery` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Readmitted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q EQ5D Index Profile` <dbl> 12112, 11111, 11111~
## $ `Post-Op Q EQ5D Index` <dbl> 0.744, 1.000, 1.000~
## $ `Knee Replacement EQ 5D Index Post-Op Q Predicted` <dbl> 0.6621424, 0.740953~
## $ `Pre-Op Q EQ VAS` <dbl> 85, 50, 46, 60, 60,~
## $ `Post-Op Q EQ VAS` <dbl> 60, 100, 90, 95, 90~
## $ `Knee Replacement EQ VAS_Post-Op Q Predicted` <dbl> 75.63073, 66.38606,~
```

```
## $ `Knee Replacement Pre-Op Q Pain` <dbl> 1, 1, 0, 0, 1, 1, 1~
## $ `Knee Replacement Pre-Op Q Night Pain` <dbl> 1, 2, 2, 2, 2, 3, 0~
## $ `Knee Replacement Pre-Op Q Washing` <dbl> 3, 4, 3, 4, 3, 3, 3~
## $ `Knee Replacement Pre-Op Q Transport` <dbl> 1, 4, 2, 4, 2, 2, 2~
## $ `Knee Replacement Pre-Op Q Walking` <dbl> 3, 3, 2, 3, 3, 1, 2~
## $ `Knee Replacement Pre-Op Q Standing` <dbl> 3, 2, 2, 3, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Limping` <dbl> 0, 3, 0, 0, 3, 1, 1~
## $ `Knee Replacement Pre-Op Q Kneeling` <dbl> 0, 0, 0, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Work` <dbl> 2, 2, 1, 3, 2, 1, 2~
## $ `Knee Replacement Pre-Op Q Confidence` <dbl> 3, 3, 3, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Shopping` <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs` <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score` <dbl> 20, 32, 17, 27, 27, ~
## $ `Knee Replacement Post-Op Q Pain` <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain` <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport` <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing` <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling` <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work` <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping` <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs` <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score` <dbl> 37, 46, 42, 47, 44, ~
## $ `Knee Replacement OKS Post-Op Q Predicted` <dbl> 34.02619, 36.88715, ~
```

```
# glimpseData <- kneeData[, c("Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
# "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
# "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index")]
# kable(glimpseData, format = "latex", booktabs = TRUE, longtable = TRUE,
# caption="Rate of Change within an Individual Health Board Against its Moving Average") %>%
# kable_styling(latex_options = c("striped", "hold_position")) %>%
# row_spec(0, bold = TRUE) %>%
# column_spec(1, "2cm") %>%
# row_spec(15, bold = TRUE) %>%
# kableExtra::landscape()

# set_flextable_defaults(
# font.family = "Arial", font.size = 10,
# border.color = "gray", big.mark = ""
# )

# ft <- flextable(head(kneeData), col_keys = c(
# "Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
# "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
# "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index"
# )) |> bold(part = "header")
# set_table_properties(ft, width=.5, layout = "autofit")
```

Dataset description

This dataset uses data from NHS England regarding PROMS for knee replacement surgery. The way in which this data has been gathered is to collect self-reported symptoms/QOL metrics through two standard procedures. One survey is the EQ-5DTM Index designed to evaluate the QOL in terms of doing daily tasks, such as (cite NHS PROMS): * Ability to pursue usual activities * Current experience of anxiety and/or depression, if any * Current experience of pain and discomfort if any * Mobility * Ability to wash and dress themselves The closer the EQ5 value is to 1, the better the patients QOL.

While the EQ5 focuses on day-to-day tasks the EQ-VAS focuses on pre/post operative outcomes through a visual thermometer scale ranging from 0 to 100. This scale is designed to aid patients in describing how they feel post operation about the results and if their problems pre operation have been addressed with the surgery.

Visually inspecting the database, we see that Age Band and Gender have many missing values. Alongside this, there are many missing values for co-morbidities. As instructed all missing values for co-morbidities denoted as 9 will be taken as a no value. However, since Age Band and Gender typically are important factors (especially as confounders), these values will be imputed. Since there are a possible 77 variables to provide summary statistics on, only specific columns relevant to the following questions below will be considered.

```
kneeData |> tbl_summary(include = c(
  "Age Band", "Gender", "Knee Replacement Post-Op Q Pain",
  "Post-Op Q EQ5D Index", "Pre-Op Q Previous Surgery", "Post-Op Q Assisted By"
))
```

Suitable machine learning algorithm for three questions:

1. Before the operation, can we estimate the post-operative EQ5D index for a patient? The EQ5 index is provided as a float value between 0 and 1. Due to its continuous nature a suitable supervised model to answer this question would be a linear regression model. Since we wish to predict a continuous variable against possible predictor variables pre/post-surgery, we would have to find correlations between an increasing EQ5 index value and variables in the dataset. There are 2 possible approaches to answering this question: a simple linear regression model and a random forest for regression. Both models won't be made, however, assumptions on the dataset will guide the choice of model; if the dataset is found to have a non-linear relationship or standard test to check for normality does not hold true, a random forest will be used. A linear regression model is simpler to run and provides better interpretability of the model, however if assumptions do not hold, a random forest for regression must be used. This question will be used to complete the report.

2. Before the operation, can we predict how much pain a patient will have after the operation? Since pain is measured as ordinal categories, a model designed for classification would fit this specification. Referencing (mention table figure) we see that 25% of data points are missing values for the questions regarding pain. If we were to use an unsupervised model it may not capture the true relationship between predictors and the outcome variable due to the vast number of unlabelled data points for a significant predictor. To tackle this issue of missing data, a semi-supervised model such as a semi-supervised support vector machine can overcome the missing data barrier. The semi-supervised SVM would use a mixture of labelled and unlabelled data to create decision boundaries for the different pain categories, and predict the positioning of a data point to its relative class given the presence of certain predictor variables. The level of computation a semi-supervised SVM has may be detrimental to the model's overall performance, but coupled with the fact that there are many other missing values in other questions regarding the QOL status of the patient, imputation on specific question would be difficult to complete without clinical knowledge to inform the possible current state of the patient. Therefore, the increased computational power for the model may be beneficial when trying to address the gaps in data.

3. Before the operation, can we calculate how many patients have had previous surgery?

Here we can use supervised/semi-supervised learning methods to calculate the numerical value of the patients

Characteristic	N = 4,528 ^I
Age Band	
*	1,603 (35%)
50 to 59	164 (3.6%)
60 to 69	1,034 (23%)
70 to 79	1,511 (33%)
80 to 89	216 (4.8%)
Gender	
*	1,603 (35%)
1	1,326 (29%)
2	1,599 (35%)
Knee Replacement Post-Op Q Pain	
0	116 (2.6%)
1	773 (17%)
2	974 (22%)
3	1,529 (34%)
4	1,136 (25%)
Post-Op Q EQ5D Index	0.76 (0.69, 1.00)
Pre-Op Q Previous Surgery	
1	437 (9.7%)
2	4,060 (90%)
9	31 (0.7%)
Post-Op Q Assisted By	
1	252 (5.6%)
2	18 (0.4%)
3	6 (0.1%)
4	9 (0.2%)
5	2 (<0.1%)
6	9 (0.2%)
9	4,232 (93%)

^In (%); Median (Q1, Q3)

that have had the previous surgery. The results is binary wether they have or have not had it

Model building to answer chosen question

1. Data splitting

Before the data is split into relevant train/test splits. The issue of Age Band and Gender both missing 35% of their data needs to be a handled. The following imputation is simple and doesn't use external questions or factors to inform decision making. Age Band is recoded into group numbers and the median of the group numbers is taken. Then the median value is then imputed to the missing data.

Imputation of the Gender column is slightly more complex, with the method being a k Nearest Neighbour (KNN) approach. This approach tries to maintain the randomness of the gender data points, (reference table) looking at the table there are 6% more females that get the surgery than males. Therefore, using a simple

50-50 or a median split would lose the close difference in value between the different genders. The KNN maintain the nuisance in this split. The following table summarises the changes in the data values after imputation, the changes are as follows * Age Band - Median = Category 3. Increase of 35% or 1,603 patients in category 3 * Gender - Male: Increase of 693 cases (+26%), Female: Increase of 910 cases (+20%)

Replacing missing values was only complete for age and gender since these values can be more easily accounted for in comparison to missing questionnaire values. Replacing missing questionnaire values may have serious consequences to the predictions of the model, creating false equivalences. With certain predictors given more importance due to the inputted data. Removing these data values was not an option, since we would be losing up to 80% of the questionnaire data with some questions (highlight table). To ensure that the integrity of the patient reported outcomes remain the same, more emphasis will be placed on the models to handle missing data such as a random forest (RF) model for regression. RF models don't need imputation of missing values and they don't suffer from underperforming from the lack of labelled data points. This is due to their automatic handling of missing data through surrogate splits. The randomForest package in R states in their documentation that imputation and handling of missing values will only take place for values coded as NA, since the missing values in this dataset are coded as 9, these will be changed.

As per the assignment specifications, 9 will be handled as "No" for any columns pertaining to a previous diagnosis of condition. 9 will be recoded to 2 for these answers. The randomForest package comes with its own built in imputation algorithm, based on first using a median/mean as a baseline to then create its own forests based on proximity data points. Values missing from the questionnaire will be replaced with the calculated values of the random forest imputation algorithm.

```
#Recode age bands into numeric groupings for an easier interpretation of the median
kneeData["Age Band"][kneeData["Age Band"] == "50 to 59"] <- "1"
kneeData["Age Band"][kneeData["Age Band"] == "60 to 69"] <- "2"
kneeData["Age Band"][kneeData["Age Band"] == "70 to 79"] <- "3"
kneeData["Age Band"][kneeData["Age Band"] == "80 to 89"] <- "4"
kneeData$"Age Band" <- as.numeric(as.character(kneeData$"Age Band"))

#Impute median values into the missing age data
kneeData["Age Band"][is.na(kneeData["Age Band"])] <- median(kneeData$"Age Band", na.rm = TRUE)
kneeData[kneeData == "*"] <- NA

#Call a KNN function designed to split the missing values comparing only to the nearest one neighbour
imputedGender <- kNN(kneeData, k = 1)
kneeData$Gender <- imputedGender$Gender
kneeData$Gender <- as.numeric(kneeData$Gender)

#Functions to replace missing values to specified values
nineToNo <- function(x){
  ifelse(x==9,2,x)
}

#Note: Imputation for the random forest package can only read NA values, all 9 values have to be recoded
nineToNA <- function(x){
  ifelse(x==9,NA,x)
}
colsToNo <- c(
  "Heart Disease", "High Bp", "Stroke", "Circulation", "Lung Disease",
  "Diabetes", "Kidney Disease", "Nervous System", "Liver Disease",
  "Cancer", "Depression", "Arthritis"
)

#Function calls to recode missing values
```

Characteristic	N = 4,528 ^I
Age Band	
1	164 (3.6%)
2	1,034 (23%)
3	3,114 (69%)
4	216 (4.8%)
Gender	
1	2,019 (45%)
2	2,509 (55%)
Heart Disease	
1	369 (8.1%)
2	4,159 (92%)
Post-Op Q Assisted By	
1	4,484 (99%)
2	18 (0.4%)
3	6 (0.1%)
4	9 (0.2%)
5	2 (<0.1%)
6	9 (0.2%)

^In (%)

```

setDT(kneeData)[, (colsToNo) := lapply(.SD, nineToNo), .SDcols=colsToNo]
kneeData[, (setdiff(names(kneeData), "Post-Op Q EQ5D Index")) := lapply(.SD, nineToNA),
  .SDcols = setdiff(names(kneeData), "Post-Op Q EQ5D Index")]

#Call function to handle missing values for questionnaire questions in dataset
kneeData <- na.roughfix(kneeData)

kneeData |> tbl_summary(
  include = c("Age Band", "Gender", "Heart Disease", "Post-Op Q Assisted By"
))

#Dataset is now split into training and test splits with
#70% of current data in training with 30% reserved for testing
kneeSplit <- kneeData %>%
  initial_split(prop = 0.7, strata = "Post-Op Q EQ5D Index")
kneeTrain <- training(kneeSplit)
kneeTest <- testing(kneeSplit)

glimpse(kneeData)

## Rows: 4,528
## Columns: 77
## $ `Age Band`                <dbl> 3, 3, 3, 3, 3, 3, 3~
## $ Gender                    <dbl> 2, 1, 2, 1, 2, 2, 2~
## $ `Pre-Op Q Assisted`       <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By`    <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period` <dbl> 2, 4, 4, 2, 4, 2, 2~

```

## \$ `Pre-Op Q Previous Surgery`	<dbl> 2, 1, 2, 2, 2, 2, 2~
## \$ `Pre-Op Q Living Arrangements`	<dbl> 2, 2, 1, 1, 1, 1, 1~
## \$ `Pre-Op Q Disability`	<dbl> 1, 2, 2, 2, 1, 2, 2~
## \$ `Heart Disease`	<dbl> 2, 2, 2, 2, 2, 1, 2~
## \$ `High Bp`	<dbl> 1, 2, 2, 1, 1, 1, 2~
## \$ Stroke	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ Circulation	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Lung Disease`	<dbl> 2, 2, 2, 1, 2, 2, 2~
## \$ Diabetes	<dbl> 2, 2, 2, 1, 2, 2, 2~
## \$ `Kidney Disease`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Nervous System`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Liver Disease`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ Cancer	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ Depression	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ Arthritis	<dbl> 2, 1, 1, 1, 1, 1, 1~
## \$ `Pre-Op Q Mobility`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Pre-Op Q Self-Care`	<dbl> 1, 1, 1, 1, 1, 1, 1~
## \$ `Pre-Op Q Activity`	<dbl> 3, 2, 3, 2, 1, 2, 2~
## \$ `Pre-Op Q Discomfort`	<dbl> 2, 2, 3, 2, 1, 2, 2~
## \$ `Pre-Op Q Anxiety`	<dbl> 1, 1, 2, 2, 1, 2, 1~
## \$ `Pre-Op Q EQ5D Index Profile`	<dbl> 21321, 21221, 21332~
## \$ `Pre-Op Q EQ5D Index`	<dbl> 0.364, 0.691, 0.030~
## \$ `Post-Op Q Assisted`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Post-Op Q Assisted By`	<dbl> 1, 1, 1, 1, 1, 1, 1~
## \$ `Post-Op Q Living Arrangements`	<dbl> 2, 2, 1, 1, 1, 1, 1~
## \$ `Post-Op Q Disability`	<dbl> 1, 2, 2, 2, 2, 1, 2~
## \$ `Post-Op Q Mobility`	<dbl> 1, 1, 1, 1, 1, 2, 1~
## \$ `Post-Op Q Self-Care`	<dbl> 2, 1, 1, 1, 1, 2, 1~
## \$ `Post-Op Q Activity`	<dbl> 1, 1, 1, 1, 1, 2, 1~
## \$ `Post-Op Q Discomfort`	<dbl> 1, 1, 1, 1, 1, 2, 2~
## \$ `Post-Op Q Anxiety`	<dbl> 2, 1, 1, 1, 1, 1, 1~
## \$ `Post-Op Q Satisfaction`	<dbl> 2, 3, 1, 2, 1, 4, 3~
## \$ `Post-Op Q Success`	<dbl> 1, 1, 1, 1, 1, 4, 1~
## \$ `Post-Op Q Allergy`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Post-Op Q Bleeding`	<dbl> 2, 2, 2, 2, 1, 2, 2~
## \$ `Post-Op Q Wound`	<dbl> 1, 2, 2, 2, 1, 2, 2~
## \$ `Post-Op Q Urine`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Post-Op Q Further Surgery`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Post-Op Q Readmitted`	<dbl> 2, 2, 2, 2, 2, 2, 2~
## \$ `Post-Op Q EQ5D Index Profile`	<dbl> 12112, 11111, 11111~
## \$ `Post-Op Q EQ5D Index`	<dbl> 0.744, 1.000, 1.000~
## \$ `Knee Replacement EQ 5D Index Post-Op Q Predicted`	<dbl> 0.6621424, 0.740953~
## \$ `Pre-Op Q EQ VAS`	<dbl> 85, 50, 46, 60, 60, ~
## \$ `Post-Op Q EQ VAS`	<dbl> 60, 100, 90, 95, 90~
## \$ `Knee Replacement EQ VAS_Post-Op Q Predicted`	<dbl> 75.63073, 66.38606, ~
## \$ `Knee Replacement Pre-Op Q Pain`	<dbl> 1, 1, 0, 0, 1, 1, 1~
## \$ `Knee Replacement Pre-Op Q Night Pain`	<dbl> 1, 2, 2, 2, 2, 3, 0~
## \$ `Knee Replacement Pre-Op Q Washing`	<dbl> 3, 4, 3, 4, 3, 3, 3~
## \$ `Knee Replacement Pre-Op Q Transport`	<dbl> 1, 4, 2, 4, 2, 2, 2~
## \$ `Knee Replacement Pre-Op Q Walking`	<dbl> 3, 3, 2, 3, 3, 1, 2~
## \$ `Knee Replacement Pre-Op Q Standing`	<dbl> 3, 2, 2, 3, 3, 2, 2~
## \$ `Knee Replacement Pre-Op Q Limping`	<dbl> 0, 3, 0, 0, 3, 1, 1~
## \$ `Knee Replacement Pre-Op Q Kneeling`	<dbl> 0, 0, 0, 2, 1, 1, 2~
## \$ `Knee Replacement Pre-Op Q Work`	<dbl> 2, 2, 1, 3, 2, 1, 2~


```
## $ `Knee Replacement Pre-Op Q Confidence`      <dbl> 3, 3, 3, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Shopping`         <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs`           <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score`            <dbl> 20, 32, 17, 27, 27,~
## $ `Knee Replacement Post-Op Q Pain`            <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain`       <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing`          <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport`        <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking`          <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing`         <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping`          <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling`         <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work`            <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence`       <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping`         <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs`           <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score`            <dbl> 37, 46, 42, 47, 44,~
## $ `Knee Replacement OKS Post-Op Q Predicted`    <dbl> 34.02619, 36.88715,~
```

```
print(colnames(kneeData))
```

```
## [1] "Age Band"
## [2] "Gender"
## [3] "Pre-Op Q Assisted"
## [4] "Pre-Op Q Assisted By"
## [5] "Pre-Op Q Symptom Period"
## [6] "Pre-Op Q Previous Surgery"
## [7] "Pre-Op Q Living Arrangements"
## [8] "Pre-Op Q Disability"
## [9] "Heart Disease"
## [10] "High Bp"
## [11] "Stroke"
## [12] "Circulation"
## [13] "Lung Disease"
## [14] "Diabetes"
## [15] "Kidney Disease"
## [16] "Nervous System"
## [17] "Liver Disease"
## [18] "Cancer"
## [19] "Depression"
## [20] "Arthritis"
## [21] "Pre-Op Q Mobility"
## [22] "Pre-Op Q Self-Care"
## [23] "Pre-Op Q Activity"
## [24] "Pre-Op Q Discomfort"
## [25] "Pre-Op Q Anxiety"
## [26] "Pre-Op Q EQ5D Index Profile"
## [27] "Pre-Op Q EQ5D Index"
## [28] "Post-Op Q Assisted"
## [29] "Post-Op Q Assisted By"
## [30] "Post-Op Q Living Arrangements"
## [31] "Post-Op Q Disability"
## [32] "Post-Op Q Mobility"
## [33] "Post-Op Q Self-Care"
## [34] "Post-Op Q Activity"
```

```

## [35] "Post-Op Q Discomfort"
## [36] "Post-Op Q Anxiety"
## [37] "Post-Op Q Satisfaction"
## [38] "Post-Op Q Sucess"
## [39] "Post-Op Q Allergy"
## [40] "Post-Op Q Bleeding"
## [41] "Post-Op Q Wound"
## [42] "Post-Op Q Urine"
## [43] "Post-Op Q Further Surgery"
## [44] "Post-Op Q Readmitted"
## [45] "Post-Op Q EQ5D Index Profile"
## [46] "Post-Op Q EQ5D Index"
## [47] "Knee Replacement EQ 5D Index Post-Op Q Predicted"
## [48] "Pre-Op Q EQ VAS"
## [49] "Post-Op Q EQ VAS"
## [50] "Knee Replacement EQ VAS_Post-Op Q Predicted"
## [51] "Knee Replacement Pre-Op Q Pain"
## [52] "Knee Replacement Pre-Op Q Night Pain"
## [53] "Knee Replacement Pre-Op Q Washing"
## [54] "Knee Replacement Pre-Op Q Transport"
## [55] "Knee Replacement Pre-Op Q Walking"
## [56] "Knee Replacement Pre-Op Q Standing"
## [57] "Knee Replacement Pre-Op Q Limping"
## [58] "Knee Replacement Pre-Op Q Kneeling"
## [59] "Knee Replacement Pre-Op Q Work"
## [60] "Knee Replacement Pre-Op Q Confidence"
## [61] "Knee Replacement Pre-Op Q Shopping"
## [62] "Knee Replacement Pre-Op Q Stairs"
## [63] "Knee Replacement Pre-Op Q Score"
## [64] "Knee Replacement Post-Op Q Pain"
## [65] "Knee Replacement Post-Op Q Night Pain"
## [66] "Knee Replacement Post-Op Q Washing"
## [67] "Knee Replacement Post-Op Q Transport"
## [68] "Knee Replacement Post-Op Q Walking"
## [69] "Knee Replacement Post-Op Q Standing"
## [70] "Knee Replacement Post-Op Q Limping"
## [71] "Knee Replacement Post-Op Q Kneeling"
## [72] "Knee Replacement Post-Op Q Work"
## [73] "Knee Replacement Post-Op Q Confidence"
## [74] "Knee Replacement Post-Op Q Shopping"
## [75] "Knee Replacement Post-Op Q Stairs"
## [76] "Knee Replacement Post-Op Q Score"
## [77] "Knee Replacement OKS Post-Op Q Predicted"

```

2. Selection and preprocessing of predictors

The first step to select important predictors to the model is to perform correlation analysis to identify predictors. There are 77 possible predictors in this dataset visually representing all the possible predictor variables made the correlation plot difficult to interpret, therefore a selection of predictor variables have been selected from the complete correlation plot (ref Appendix) only if they had a value ± 0.5 associated with any of the variables. This is a purely numeric based approach, however, there may be certain variables that hold clinical significance and wish to be included in the model to evaluate their relationship.

*#Trailing spaces were present in some column names, these needed to be stripped.
 #All spaces in the columns were replaced with a . character*

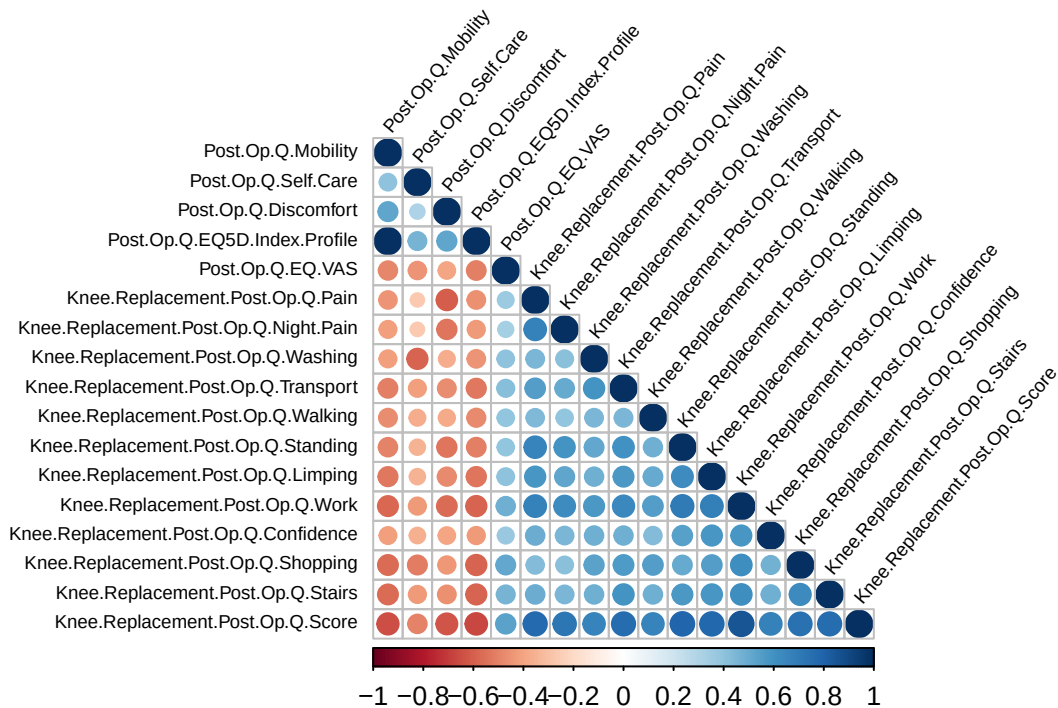
```

colnames(kneeTrain) <- make.names(colnames(kneeTrain))
colnames(kneeTest) <- make.names(colnames(kneeTest))

kneeCorColumns <- kneeTrain[, c(
  "Post.Op.Q.Mobility", "Post.Op.Q.Self.Care", "Post.Op.Q.Discomfort",
  "Post.Op.Q.EQ5D.Index.Profile", "Post.Op.Q.EQ.VAS", "Knee.Replacement.Post.Op.Q.Pain",
  "Knee.Replacement.Post.Op.Q.Night.Pain", "Knee.Replacement.Post.Op.Q.Washing",
  "Knee.Replacement.Post.Op.Q.Transport", "Knee.Replacement.Post.Op.Q.Walking",
  "Knee.Replacement.Post.Op.Q.Standing", "Knee.Replacement.Post.Op.Q.Limping",
  "Knee.Replacement.Post.Op.Q.Work", "Knee.Replacement.Post.Op.Q.Confidence",
  "Knee.Replacement.Post.Op.Q.Shopping", "Knee.Replacement.Post.Op.Q.Stairs",
  "Knee.Replacement.Post.Op.Q.Score"
)]

kneeCorr <- cor(kneeCorColumns)
corrplot(kneeCorr,
  type = "lower", tl.srt = 50,
  tl.col = "black", number.cex = 0.5, tl.cex = 0.6
)

```



```

# Create a recipe with significant predictor variables
# Recipe used for more readable code when fitting the model, assignment specificatoin mention not fit
# In the case of multiple models being fitted this recipe can be used multiple times to the parameter

```

```

kneeModelRec <- kneeTrain %>%
  recipe(Post.Op.Q.EQ5D.Index ~

```

```

Post.Op.Q.Mobility +
Post.Op.Q.Self.Care +
Post.Op.Q.Discomfort +
Post.Op.Q.EQ5D.Index.Profile +
Post.Op.Q.EQ.VAS +
Knee.Replacement.Post.Op.Q.Pain +
Knee.Replacement.Post.Op.Q.Night.Pain +
Knee.Replacement.Post.Op.Q.Washing +
Knee.Replacement.Post.Op.Q.Transport +
Knee.Replacement.Post.Op.Q.Walking +
Knee.Replacement.Post.Op.Q.Standing +
Knee.Replacement.Post.Op.Q.Limping +
Knee.Replacement.Post.Op.Q.Work +
Knee.Replacement.Post.Op.Q.Confidence +
Knee.Replacement.Post.Op.Q.Shopping +
Knee.Replacement.Post.Op.Q.Stairs +
Knee.Replacement.Post.Op.Q.Score) %>%
step_zv(all_predictors()) %>%
step_corr(all_predictors())

```

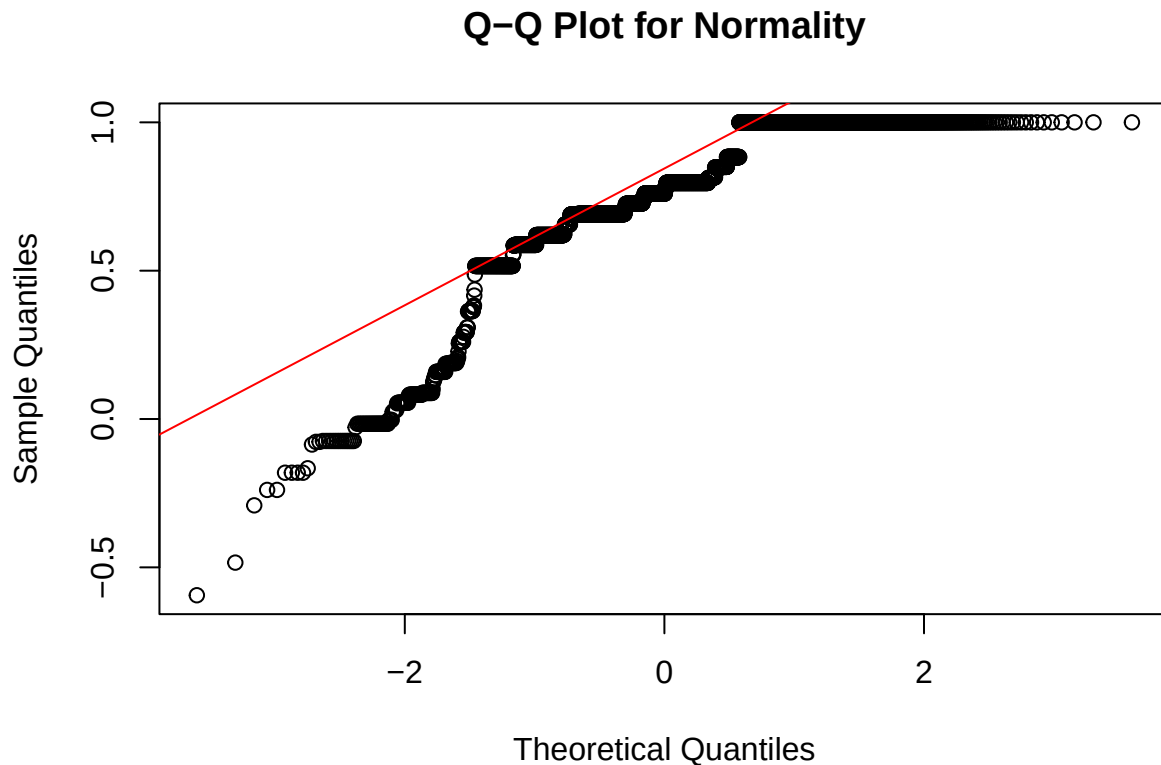
3. Model specification and training

Missing values in the dataset have been addressed alongside identifying important predictor variables that have influence on the dependent variable. There still lies the issue of using a linear regression model or a random forest. Due to its simpler computation and interpretability linear models are preferred but a series of assumptions need to hold true, in order to use this model. The following code runs a simple Q-Q test that checks for normality within the data.

```

qqnorm(kneeTrain$Post.Op.Q.EQ5D.Index, main = "Q-Q Plot for Normality")
qqline(kneeTrain$Post.Op.Q.EQ5D.Index, col="red")

```



As shown in the Q-Q plot, the data in terms of the dependent variable does not hold the assumption of normality. Since this assumption is unmet, a random forest plot will be used. Random forest plots are capable of handling regression problems with non-linear properties.

Calling the Random Forest Model

The following code depicts the RF model being called for the purposes of fitting a regression model. RF works based on ensemble learning, where multiple decision trees are used to reach a consensus on the. In this problem specification the tree first starts off with the full complete training dataset and creates multiple branches of decision trees, giving each decision tree its own sample size (with replacement) of the original training dataset. This process is repeated for every decision tree in the RF as shown in the workflow below. These decision trees all contain a unique subset of the training data and reach a predicted value for the Post EQ5 Index based on their own subset of the data. It's important to note that each subset may not have the same types of significant variables as found in the correlation plot when predicting the value. This is an important feature, as this is how the model determines which variables are important to the model by comparing the accuracy of the trees that contain certain variables with the actual value of the Post EQ5 Index. To ensure that the trees have a collective prediction that is accurate, they go through a process known as bootstrapping designed to average out the predicted values across all the different trees. There is an optimal number of decision trees to consider that will decrease the overall variance while maintaining computational efficiency, but having a larger number of trees doesn't necessarily mean increased model performance. The model reaches a point where the increase of computation does not justify the marginal increase in the predicted value. This problem arises due to overfitting within the model, where the RF starts to learn from noise in the data, rather than making generalisable predictions based on the data. To find the optimal number of trees the model would have to go through hyper tuning tests which is outside the scope of this report, therefore the default value of decision trees built in within the randomForest package will be used. This value was not cited within the documentation.

```

#Initate random forest model
kneeRandForestMDI <- rand_forest() %>%
  set_mode("regression") %>%
  set_engine("randomForest")

# Use recipie against specifications of model
kneeWorkflowMDI <- workflow() %>%
  add_recipe(kneeModelRec) %>%
  add_model(kneeRandForestMDI)

# Fit model with training data
kneeFitModelMDI <- kneeWorkflowMDI %>%
  fit(data = kneeTrain)

#Specify a model with built in importance feature for random forest that calulates both OOB and MDI
kneeRandForestVSE <- rand_forest() %>%
  set_mode("regression") %>%
  set_engine("randomForest", importance = TRUE)

kneeWorkflowVSE <- workflow() %>%
  add_recipe(kneeModelRec) %>%
  add_model(kneeRandForestVSE)

extractRF <-
kneeFitModelVSE <- kneeWorkflowVSE %>%
  fit(data = kneeTrain)

```

Identifying Influential Variables

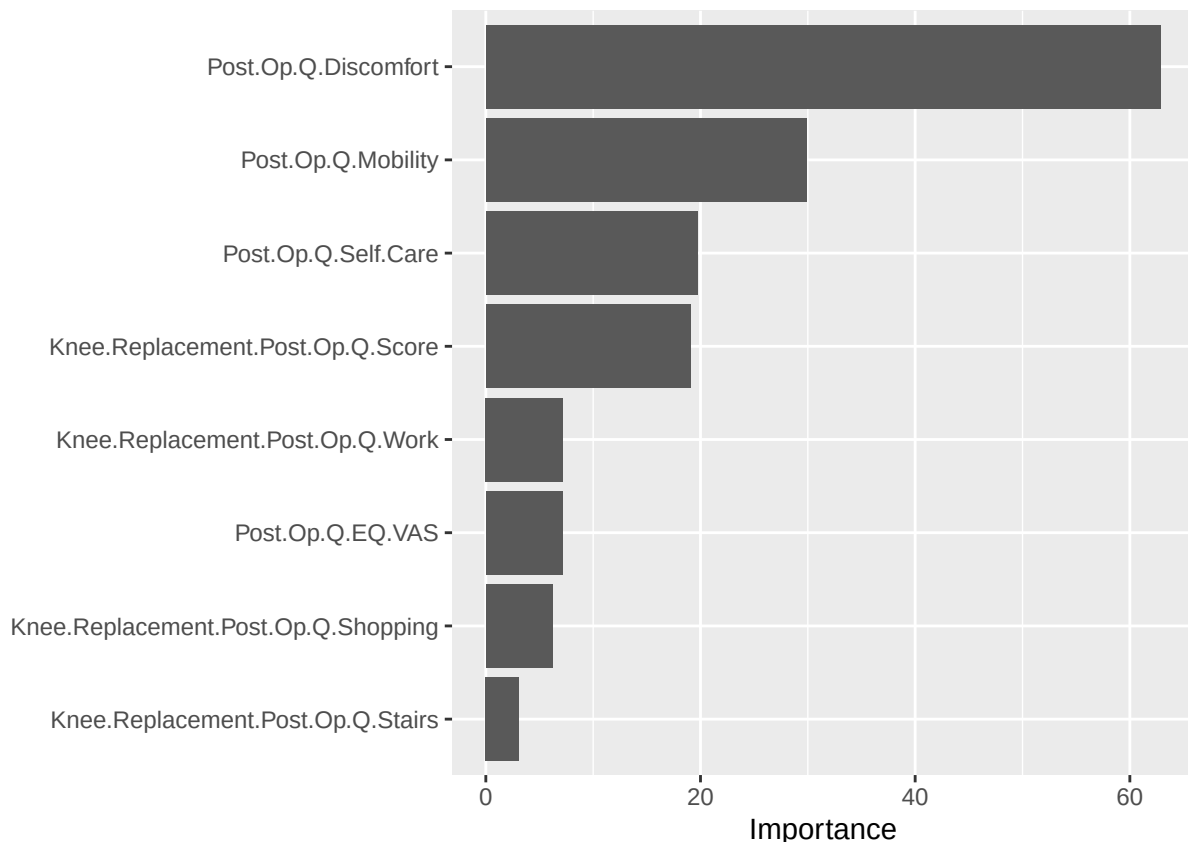
Out of the 16 possible significant variables, only 4 showed a degree of importance. In this report the threshold for importance was categorised as any value of importance greater than 20 units. The way in which the vip package places importance on variables is significant to our understanding of how variables affect the model and how they are important to the final Post EQ5 Index. The vip package uses a method known as Mean Decrease Impurity (MDI). MDI works based on calculating the total decrease in node impurity, that is, the total amount of misclassified/inaccurate values within nodes in a decision tree. The MDI is then averaged across all the trees and the predictor variables that decrease the MDI value are considered to have the most importance in the model. The documentation notes certain features that impact the validity of the results, such as: high correlation in the variables, varying scales of measurement for the predictor variables and MDI performing better in larger ensemble sizes. In the context of this report, there were some issues encountered that directly affect the way the data interacts with the vip package. Upon visual inspection there was a certain predictor “Knee Replacement Post Op Q Score” that would change between ± 8 units in the importance score, this was significant due to the threshold of an important variable being 20 units and depending on the specific run, that specific variable was considered important. Knee Replacement Post OP Q score can be assumed to be correlated to the Post Op Q Pain score, looking at the PROMS data dictionary, it states

“Each of the KR question has 5 possible responses which each are attributable 0-4 points. The KR score is derived by adding the points for each question. The maximum number of points is 48 which would signify the best outcome, the minimum being 0.”

Since the KR Post Op Q score is an aggregate variable, it will change depending on the specific variables included in the sample sizes of the decision trees in the RF model. This high correlation between Knee Post OP Q Score and the presence of other variables within a specific sample of the decision tree may indicate that this form of identifying variables is inaccurate. This could have significant impact in a clinical setting.

Providing inaccurate estimations on predictor variables would not aid clinical professionals on making well informed decisions on creating a treatment plan for patients in order to maximise health outcomes after their knee replacement surgery. Another form of identifying important variables is implemented using the Out-Of-Bag (OOB) method, where Validation Set Errors (VSE) are created for each tree. For each VSE predictors will be randomly shuffled around the trees and the variables that consistently decrease the VSE error across the decision trees will have a higher importance value.

```
kneeFitModelMDI %>%
  extract_fit_parsnip() %>%
  vip(num_features=8)
```



```
# Trained RF model needs to be extracted from tidymodels workflow in order to use it as matrix object
extractRF <- extract_fit_parsnip(kneeFitModelVSE)
MeanImportanceValues <- importance(extractRF$fit)

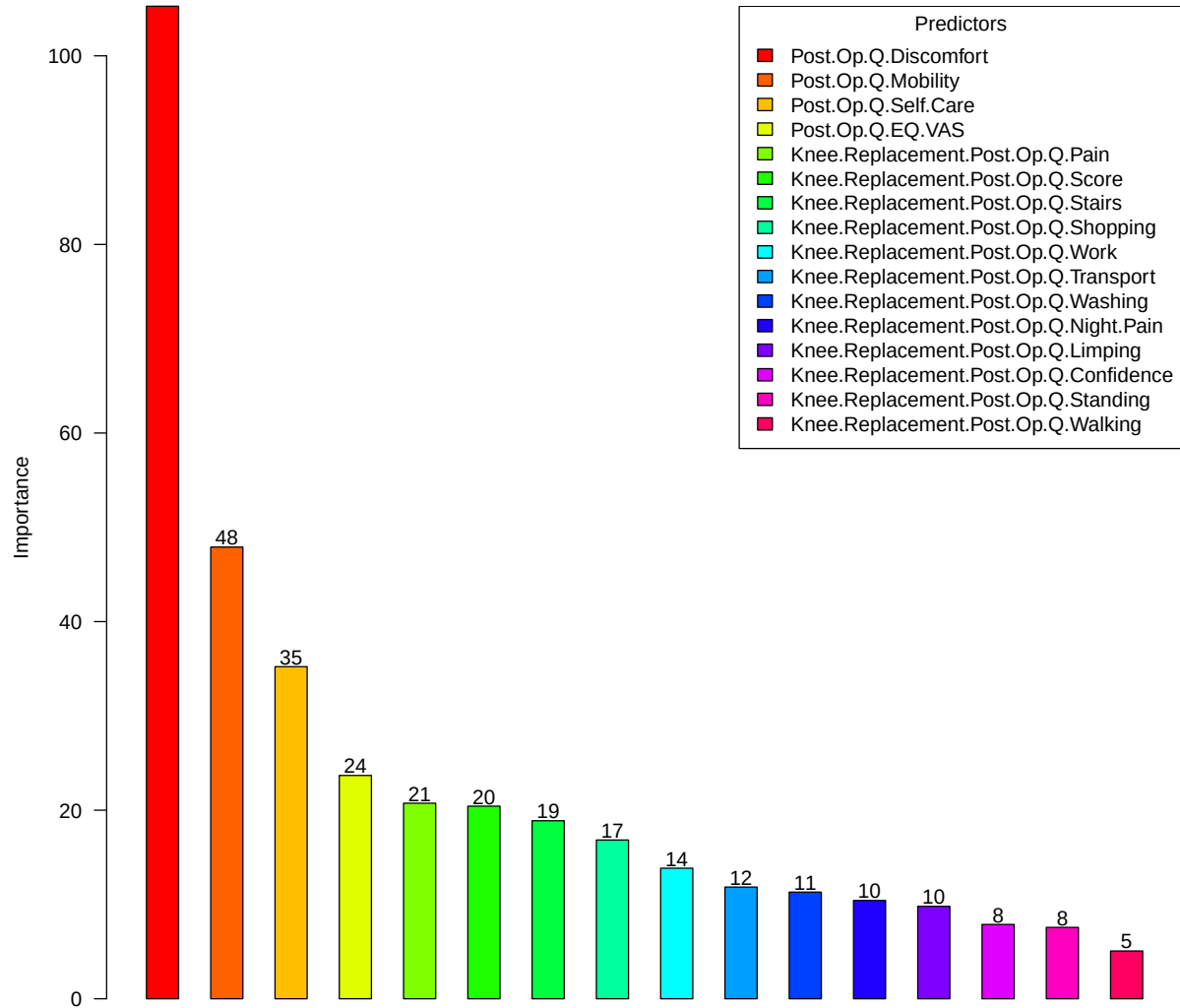
# Extract only the OOB importance values from the importance fitted RF model
MeanImportanceValuesNode <- MeanImportanceValues[, 1, drop = FALSE]

# Line taken from https://www.statology.org/sort-matrix-in-r/
sortedMeanImportanceValues <- MeanImportanceValuesNode[order(MeanImportanceValuesNode[, 1], decreasing = TRUE), ]

colors <- rainbow(nrow(MeanImportanceValuesNode))

plot<- barplot(
  t(sortedMeanImportanceValues),
  beside = TRUE,
  las = 2,
```

```
cex.names = 0.5,  
col=colors,  
xaxt="n",  
ylab="Importance"  
  
)  
text(  
  x = plot,  
  y = sortedMeanImportanceValues + 1.0,  
  labels = round(sortedMeanImportanceValues)  
)  
legend("topright",  
  legend = rownames(sortedMeanImportanceValues),  
  fill = colors,  
  title = "Predictors"  
)
```

As seen through both graphs, there are three variables that consistently show themselves as being important to the model: Post Op Q Discomfort, Post Op Q Self Care, Post Op Q Mobility. Running the second form of identifying important variables were found to have greatly differing values from the MID method. Past these 3 variables it is unclear which variables are influential to the model, these limitations will be discussed.

4. Model evaluation

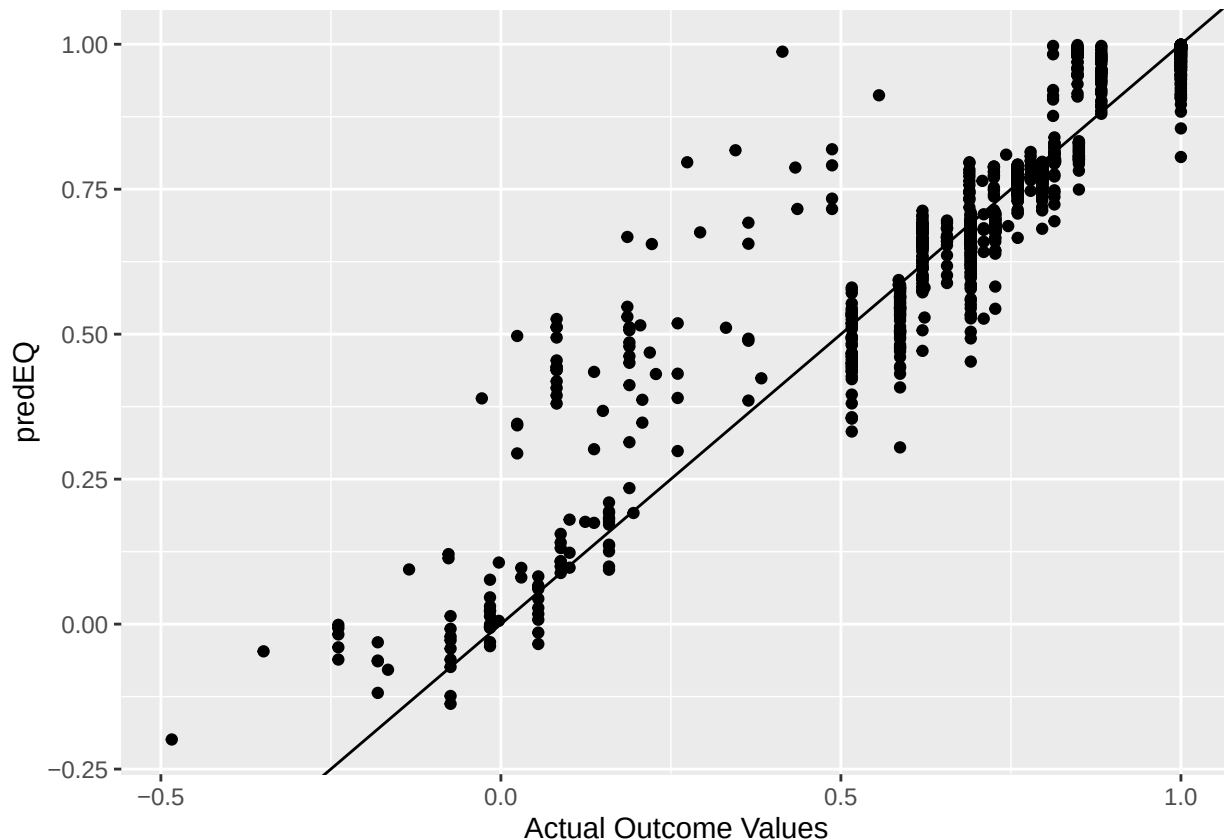
The following metrics define the performance of the model. There is no distinction between MDI or OOB when it comes to model performance since these methods only differ in the way they select variables and train/test the models with the same RF model. The model is seen to have a good overall fit with the data. The plot of actual/predicted outcome variables shows a good enough distance to indicate the model is not overfitting to the training data but seems to be underperforming in modelling the correct outcome data. Overall, the RF model is a good fit for this data with certain predictor variables being identified as significantly important variables to the model. This model may be used in aiding clinical decision making in

improving patient health outcomes after knee replacement surgery.

```
predEQValVSE <- kneeFitModelVSE %>%  
  predict(new_data = kneeTest)  
  
kneeResultVSE <- kneeTest %>%  
  bind_cols(predEQValVSE) %>%  
  rename(predEQ = .pred)  
  
metrics(kneeResultVSE, truth = Post.Op.Q.EQ5D.Index, estimate = predEQ)
```

```
## # A tibble: 3 x 3  
##   .metric .estimator .estimate  
##   <chr>   <chr>      <dbl>  
## 1 rmse    standard    0.0853  
## 2 rsq     standard    0.897  
## 3 mae     standard    0.0479
```

```
kneeResultVSE %>%  
  ggplot(aes(x = Post.Op.Q.EQ5D.Index, y = predEQ)) +  
  geom_abline(intercept = 0, slope = 1) +  
  geom_point() +  
  xlab("Actual Outcome Values")
```



```
ylab("Predicted Outcome Values")
```

```
## $y  
## [1] "Predicted Outcome Values"
```

```
##
## attr(,"class")
## [1] "labels"
```

Limitations of machine learning model

There are several limitations to this model that will be affecting the performance * Default number of decision trees being used, actual value is unknown. - Both the VIP and randomForest package document that the model performance performs better with a larger ensemble size. This could be due to the fact that there are more decision trees to make predictions on therefore the overall prediction would be more accurate - Not necessarily more decision trees equates to a more precise prediction outcome. Computational cost will outweigh the gains in how precise predicted outcome is - To address this issue, parameter hypertuning needs to be conducted to model the performance of the RF model with an increasing number of decision trees.

- High level of correlation between variables
 - Between the predictor variables there are a few variables that represent the same concept such as the EQ5 Index and the EQ VAS values. The Knee Replacement Post Op Q Pain and Post Op Q Discomfort are also highly correlated with each other as they both represent a level of pain after the surgery.
 - These metrics may have different meanings in a clinical setting that would set a precedence for different outcomes.
 - To address this issue, further clarification from the authors of the data would indicate if these variables are correlated together as hypothesised. Clinical input would also aid in the interpretation of how relevant these variables would be in the process of creating guidelines for patient outcomes.
- High levels of missing data
 - Many of the data points were missing and were imputed with basic data imputation. This may influence the model towards predicting certain outcomes that otherwise would have no relevance, creating a false equivalency.
 - Missing data can be categorised as missing at random (MAR) or missing not at random (MNAR). Due to the nature of the data being self reported and there being no incentives being not reporting data this data can be assumed to be MAR
 - More complex methods of data imputation can be used to address this issues through novel approaches directly geared towards healthcare settings where datasets exhibit a high level of cardinality.

Appendix

```
kneeCorr <- cor(kneeTrain)
corrplot(kneeCorr, method = "color", type = "lower", tl.col = "black", tl.srt = 50, number.cex = 0.5, t
```

